FINAL SUBMISSION · TEAM ERDŐS

Improving Scientific Reasoning of Vision-Language Models on JEE Advanced

A code sandbox, agentic correction, a learned answer selector and retrieval around a frozen 8B model on one GPU, tested on an exam the model has never seen.

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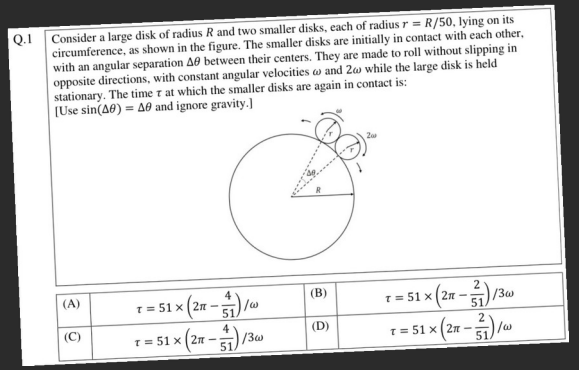
Retrieval module · code sandbox module

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Agentic correction module · learned verifier

Base paper: Mukherjee & Ghosh, mmJEE-Eval, Findings of IJCNLP-AACL 2025



Q.1 Consider a large disk of radius R and two smaller disks, each of radius $r = R/50$, lying on its circumference, as shown in the figure. The smaller disks are initially in contact with each other, with an angular separation $\Delta\theta$ between their centers. They are made to roll without slipping in opposite directions, with constant angular velocities ω and 2ω while the large disk is held stationary. The time τ at which the smaller disks are again in contact is:
[Use $\sin(\Delta\theta) = \Delta\theta$ and ignore gravity.]

(A)	$\tau = 51 \times \left(2\pi - \frac{4}{51}\right)/\omega$	(B)	$\tau = 51 \times \left(2\pi - \frac{2}{51}\right)/3\omega$
(C)	$\tau = 51 \times \left(2\pi - \frac{4}{51}\right)/3\omega$	(D)	$\tau = 51 \times \left(2\pi - \frac{2}{51}\right)/\omega$

A question from our clean test set: JEE Advanced 2026, Paper 1. The model sees only this image.

IIT BHILAI · MACHINE LEARNING

Open models still fail most JEE Advanced problems

mmJEE-Eval tests vision-language models on 1,460 JEE Advanced questions (2019–2025) in English and Hindi. Each question is an image: text, formulas, options and often a diagram.

37–45%

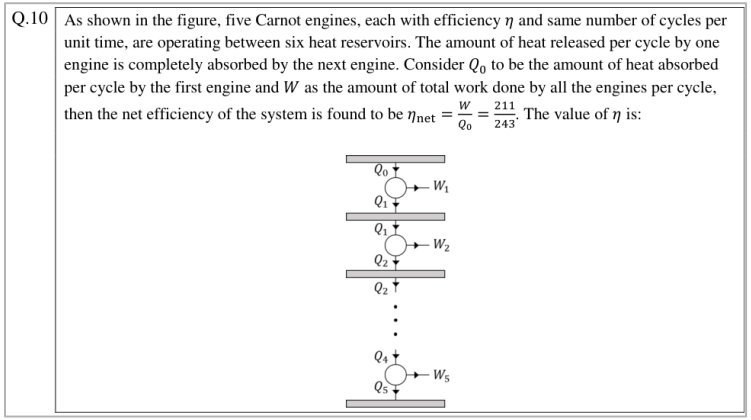
of the 2025 questions solved by open models; the best closed models solve 77–84%

44.7%

of the benchmark is numerical questions, where small calculation slips cost the whole answer

1.1–5.2%

of their own errors fixed by models, although they notice an error in 21–73% of cases



JEE Advanced 2026, Paper 1, Q.10 (numerical, with a diagram): one of the 204 questions in our clean test set.

OUR GOAL

> *Help a free, open model solve more of these questions without retraining it, and test this fairly on exams it has never seen.*

Statistics on this slide are from the base paper.

Why this problem is worth solving

01

A measured gap, not yet fixed

The base paper only diagnoses. Two of its gaps are concrete and measurable: detecting vs. fixing errors, and weak numerical accuracy.

02

Fits real hardware

Test-time methods need engineering and careful evaluation, not large-scale training. A frozen 4-bit model runs on one 12 GB lab GPU.

03

Free labels for learning

Gold answers let us mark every sampled solution right or wrong automatically: thousands of training examples at no annotation cost.

04

A fair test is possible

Exams are dated, so we can test on a paper written after the model was released, with the paper's own scoring code.

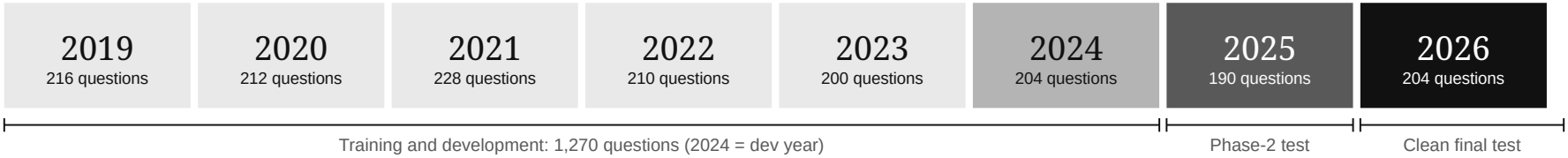
Five steps, each one prompted by the last

PHASE 1	PHASE 2	STEP 3	STEP 4	STEP 5
Proposal	**Build and test**	**Clean test**	**Better selection**	**Diagnose**
Four modules planned around a frozen model	Reproduce the paper, pilot, 7 fixes, full run on 2025	New 2026 exam, rules fixed in advance, stronger system	Six ideas, each with a pass rule set first	Where does the gain really come from?
SOP	65.3% on 2025	58.8% on unseen 2026	0 of 6 passed	pool not selection

> *Phase 2 worked, but 2025 had shaped the design, so we tested on 2026. There the selector stopped helping, so we tried better selectors. None helped, so we measured where the gain comes from.*

A fair test needs strict rules

HOW WE SPLIT THE EXAMS



1 **Same scorer**

The paper's own answer-extraction and scoring code, copied unchanged.

2 **Same compute**

Our baseline is a majority vote over plain answers with as many generated tokens as our system.

3 **Honest intervals**

English and Hindi twins are near-duplicates, so intervals treat each pair as one unit (cluster bootstrap).

4 **No leakage**

No test question or near-copy in any training data or retrieval base.

5 **Choices on old years**

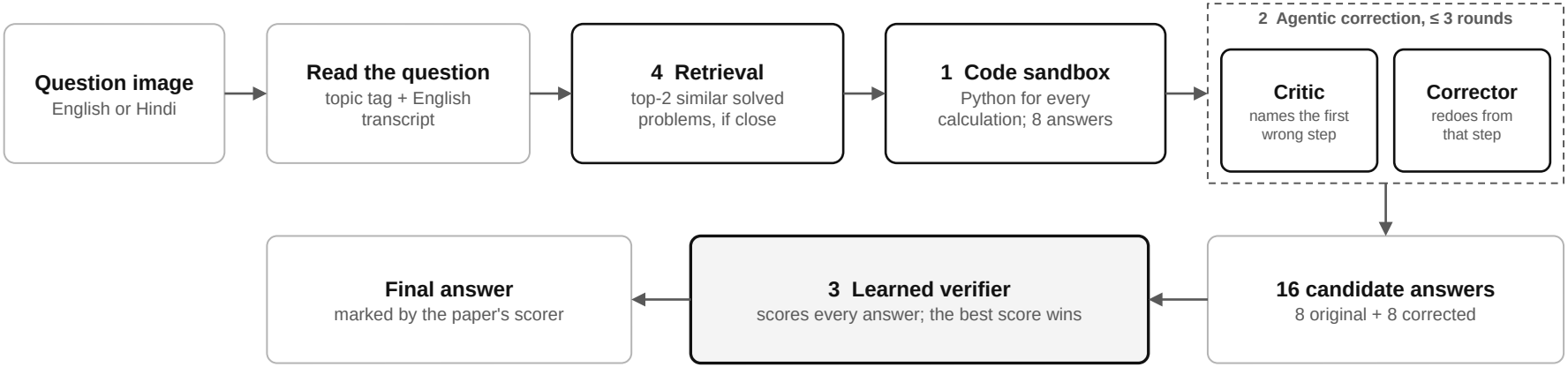
Every design choice uses 2019–2024 only; the final system is frozen before the 2026 run.

6 **Rules before runs**

Later ideas and their pass rules are committed to GitHub before they are run.

One frozen model, four modules, one trained part

ANSWERING ONE TEST QUESTION



Numbers 1–4 mark the four modules (next four slides).

PREPARED IN ADVANCE FROM 2019–2024 ONLY

Labelled answers

1,270 past questions × 8 answers, marked right or wrong by the scorer

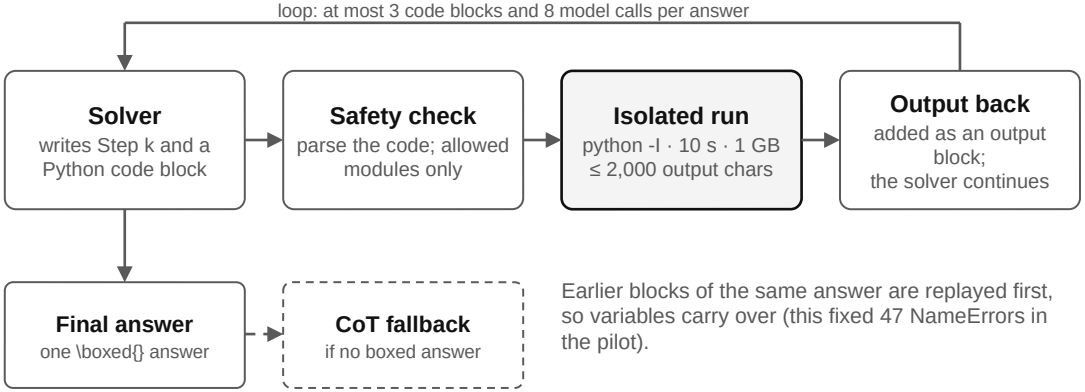
Knowledge base

verified correct solutions (+ JEEBench 2016–2018), test years removed

Verifier training

logistic regression vs. XGBoost, chosen by leave-one-year-out

Module 1 · Code makes the arithmetic exact



Earlier blocks of the same answer are replayed first, so variables carry over (this fixed 47 NameErrors in the pilot).

WHAT THE MODEL WRITES (FORMAT EXAMPLE FROM OUR PROMPT)

````
Step 1: <which law / concept applies and the equation to use>
```python
import sympy as sp
x = sp.symbols('x')
print(sp.solve(2*x - 6, x))
```
```output
[3]
```
Step 2: <interpret the printed result, check the options>   \boxed{<answer>}
````

TECHNICAL DETAILS

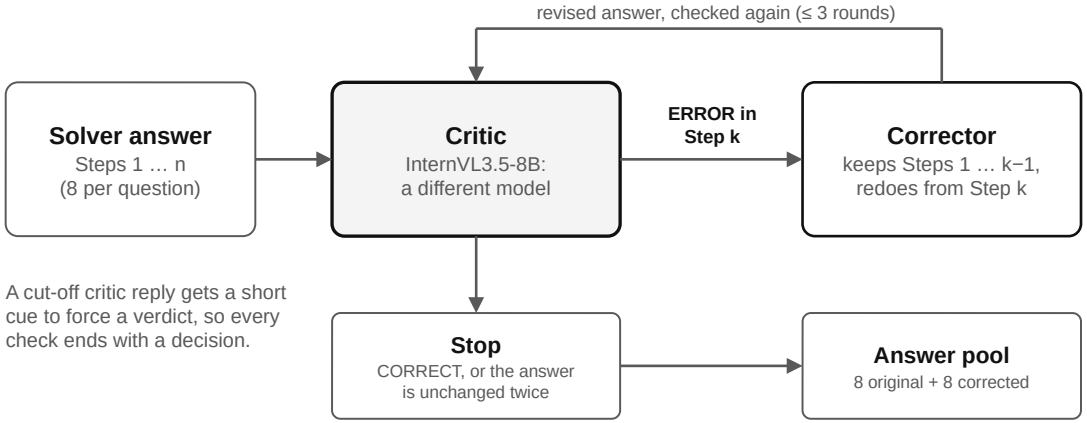
MODEL	Qwen3-VL-8B, 4-bit AWQ, frozen; 8 answers per question, temperature 0.7
ALLOWED	math, sympy, numpy, scipy, fractions, statistics, itertools …; no files, no network, no parsers
LIMITS	10 s, 1,024 MB and 2,000 output characters per block
BUDGET	4,096 tokens; the final system continues a cut-off answer once (+4,096)

RESULT (2025, PHASE 2)

40→50%
numerical questions solved

+5.0
points over one answer (p = 0.012)

Module 2 · A second model points at the first wrong step



HOW EVERY CRITIC REPLY MUST END (FROM OUR PROMPT)

```
VERDICT: CORRECT or ERROR
FIRST_ERROR_STEP: <step number, or NONE>
EXPLANATION: <one or two sentences: what is wrong and how to fix it>

If the reply is cut off, we append:
"I must stop checking now and give my decision.  VERDICT:"
```

TECHNICAL DETAILS

CRITIC	InternVL3.5-8B, 4-bit, greedy; short check, up to 1,536 tokens
WHY TWO	a different model family reduces shared blind spots
CORRECTOR	the solver itself, with the same code rules and the critic's explanation
GATE	final system: keep a fix only if the verifier score rises (δ = −0.08)

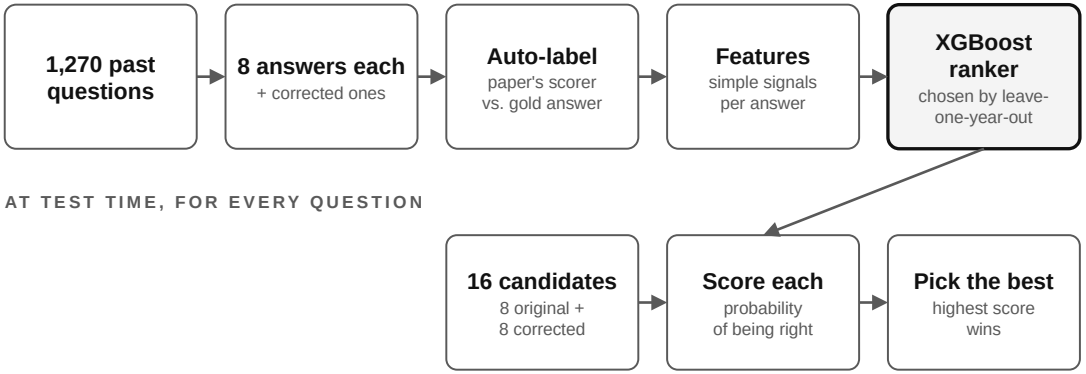
RESULT (2025, PHASE 2)

14–19%
of detected errors fixed
(paper: 1.1–5.2%)

87 / 39
wrong→right vs.
right→wrong flips

Module 3 · A small classifier picks the final answer

TRAINING, ON 2019–2024 ONLY



FEATURES (ONE ROW PER CANDIDATE ANSWER)

```
length · number of steps · code ran OK · cut off at the limit
hedging words · valid answer format
agreement with the other answers (whole pool and each half)
is the majority answer · critic flagged it · rounds · answer changed
Phase 2 also: a sentence embedding of the answer (bge-small, 32 dims)
```

TECHNICAL DETAILS

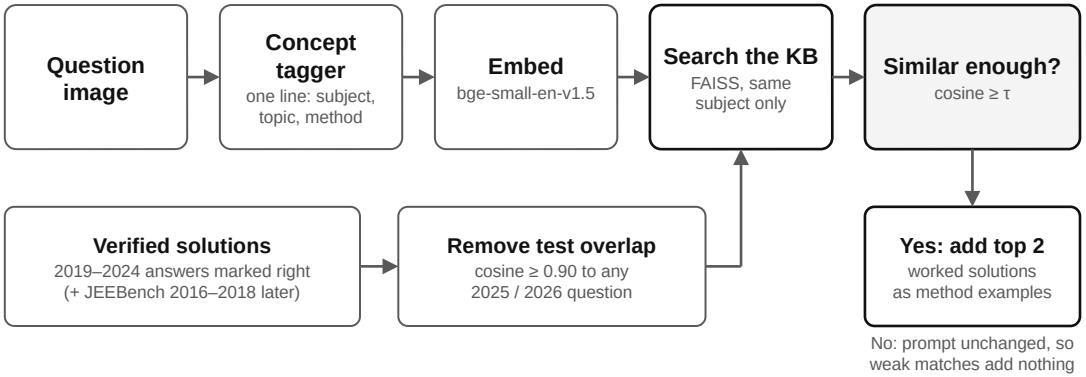
DATA	10,160 labelled answers in Phase 2 (45.5% right); 12,096 incl. corrected later
MODELS	logistic regression vs. XGBoost; final: pairwise ranker, 400 trees, depth 4
SPLIT	by exam year, so answers to one question never sit on both sides
PICK	highest score among the 16 candidates

RESULT

+11.8
points: biggest single step (2025, Phase 2)

62.0%
pick accuracy vs. 53.7% for majority vote (LOYO)

Module 4 · Solved past problems as worked examples



TAGGER OUTPUT AND EXEMPLAR HEADER (FROM OUR PROMPTS)

```
SUBJECT: <Physics/Chemistry/Mathematics> | TOPIC: <2-6 words>
| METHOD: <key concept or technique, at most 15 words>

"Below are worked solutions of earlier JEE Advanced problems that
 use a similar method. Use them only as guidance on the method.
 They are DIFFERENT problems: do not copy their numbers or answers."
```

TECHNICAL DETAILS

QUERY	the model's own tag, greedy, ≤ 160 tokens (no OCR needed)
KB	Phase 2: 454 verified entries; later + JEEBench and tag + text matching
THRESHOLD	τ tuned on old years for 70% proxy precision (Phase 2: 0.885)
COVERAGE	final recipe 11.7% vs. 8.4% of questions on old years

RESULT (2025, PHASE 2)

30 / 190
questions found a close example

30→38%
accuracy on those questions (code solver)

First, prove that our measuring tool is right

13.8%

the paper's Qwen2.5-VL-7B, run through our code (10 runs)

12.4%

the same model, as reported in the paper

The paper's value lies inside our 95% interval (10.7–17.3), so prompts, image handling and scoring match the original.

> *Our tool matches the paper, and the baseline to beat is already strong.*

OUR SOLVER BEFORE ANY MODULE

43.4%	Qwen3-VL-8B, one answer (16 runs)
53.7%	majority vote over 16 answers: the strong rival
−10 / −18	points for Hindi / for questions that need the diagram
40%	of answers cut off at 4,096 tokens

A 40-question pilot: every module adds accuracy

Pilot setup (40 test questions)	Accuracy	95% CI
Single answer (CoT)	29.5	17.8–41.6
Majority vote, same compute	35.0	20.0–52.5
+ Code sandbox	41.2	28.1–54.4
+ Correction, same-model critic	46.6	32.8–60.3
+ Correction, cross-model critic	43.1	29.7–56.2
+ Learned verifier	50.0	32.5–67.5
Full framework	**57.5**	**40.0–72.5**

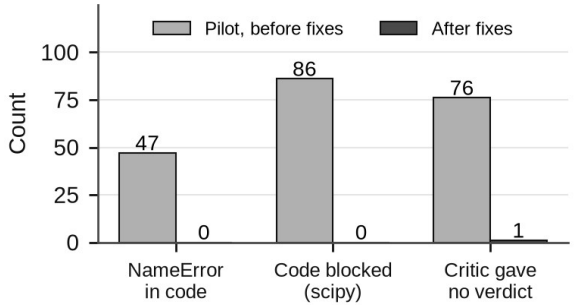
WHY A PILOT FIRST

- The full run costs about 54 GPU-hours, so mistakes are expensive.
- Full framework vs. majority vote: +22.5 points (interval +7.5 to +37.5). With 40 questions this is only a first signal.
- Its real value: reading 1,520 solution traces and 640 correction chains showed us what was broken.

> *Seven problems found and fixed before the 54-hour full run.*

Seven problems found in the pilot, fixed before the full run

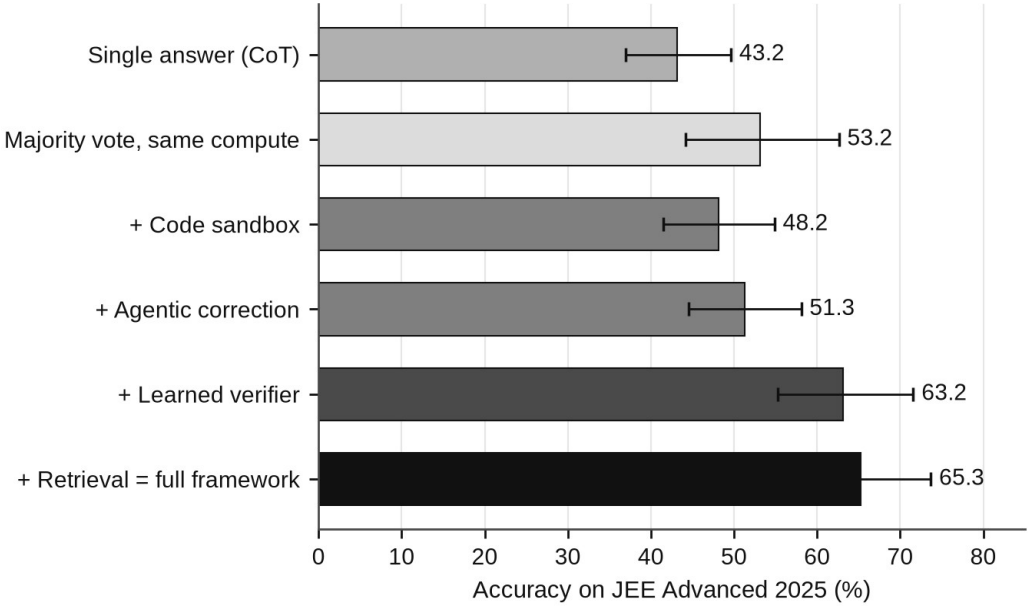
Problem	Evidence	Fix and effect
Code state lost	47 NameErrors: each block ran fresh	replay earlier blocks → 0
Library blocked	86 blocks needed SciPy	allow SciPy, no file I/O → 0
No critic verdict	76 replies cut off while re-solving	short check + forced verdict: 1 of 501 left; detection 35→56%
Many fallbacks	480 of 539 were long reasoning loops	forcing an answer tested: 21.9% vs. 30.0% → rejected
Baseline too small	system used ≈ 18 answers' tokens, only 16 drawn	draw 20 answers for a fair vote
Selector = vote	dev pick equal to majority vote	reported honestly; 8× more data
Batch aborted	one 12,303-token prompt	retry request by request



Same 40 questions, before vs. after the fixes.

A three-person code review also found 11 bugs before any GPU run, e.g. a verifier crash, a train/test feature mismatch and intervals that were too narrow.

Phase 2 on the 2025 exam: 65.3%, +12.1 over majority vote



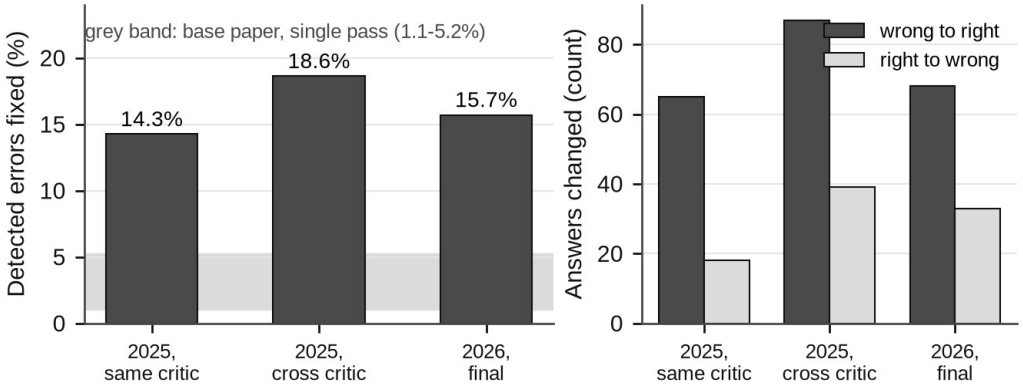
190 questions, English + Hindi. Bars show 95% intervals; each “+” row adds one module.

65.3%

full system, up from 43.2% for one answer

- +12.1 points over majority vote at equal compute (interval +4.2 to +20.5, p = 0.001).
- Numerical questions: 40.2% → 50.4% with the code sandbox.
- Learned verifier: the biggest single step, +11.8.
- Retrieval: +2.1, not significant.
- No memorisation sign: 43.9% on 2019–24 vs. 42.6% on 2025.

What Phase 2 left open



Left: detected errors that end correct after up to 3 rounds (grey band: the paper's single-pass rate). Right: answers flipped by correction.

- Our loop fixes 14–19% of the errors it detects, well above the paper's 1.1–5.2%: the critic points at one step and the solver can run code.
- But the critic also flags 19–26% of correct answers, so some right answers turn wrong.

1 **Picking**

The verifier picked 65.3%, about the same as a plain vote over the same 16 answers (65.8%). The gain came from the varied pool.

2 **Retrieval**

Only 30 of 190 questions found a close example.

3 **Budget**

36% of solutions fell back to plain reasoning, mostly at the token limit.

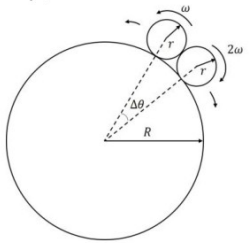
4 **Fairness**

2025 had been used for the pilot, and the model may have seen it.

A clean test: the 2026 exam, in two languages

Q.1

Consider a large disk of radius R and two smaller disks, each of radius $r = R/50$, lying on its circumference, as shown in the figure. The smaller disks are initially in contact with each other, with an angular separation $\Delta\theta$ between their centers. They are made to roll without slipping in opposite directions, with constant angular velocities ω and 2ω while the large disk is held stationary. The time τ at which the smaller disks are again in contact is:
[Use $\sin(\Delta\theta) = \Delta\theta$ and ignore gravity.]

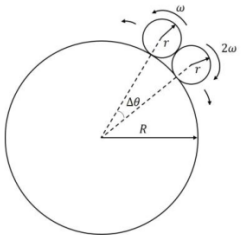


(A)	$\tau = 51 \times \left(2\pi - \frac{4}{51}\right)/\omega$	(B)	$\tau = 51 \times \left(2\pi - \frac{2}{51}\right)/3\omega$
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ENGLISH · FROM THE DATASET

Q.1

एक R त्रिज्या की बड़ी चकती (disk) की परिधि पर स्थित दो छोटी $r = R/50$ त्रिज्या की चकतियां दर्शाए चित्रानुसार स्थित हैं| प्रारंभ में छोटी चकतियाँ एक दूसरे से इस प्रकार संपर्क में हैं कि उनके केन्द्रों के बीच कोणीय दूरी $\Delta\theta$ है| छोटी चकतियों को बड़ी चकती की परिधि पर बिना फिसले विपरीत दिशाओं में अचर कोणीय वेग ω और 2ω से लुढ़काया जाता है जबकि बड़ी चकती को स्थिर रखा गया है | समय τ , जिस पर छोटी चकतियाँ पुनः संपर्क में आती है, होगा:
[गुरुत्वाकर्षण की उपेक्षा कीजिये एवं $\sin(\Delta\theta) = \Delta\theta$ का प्रयोग कीजिये]

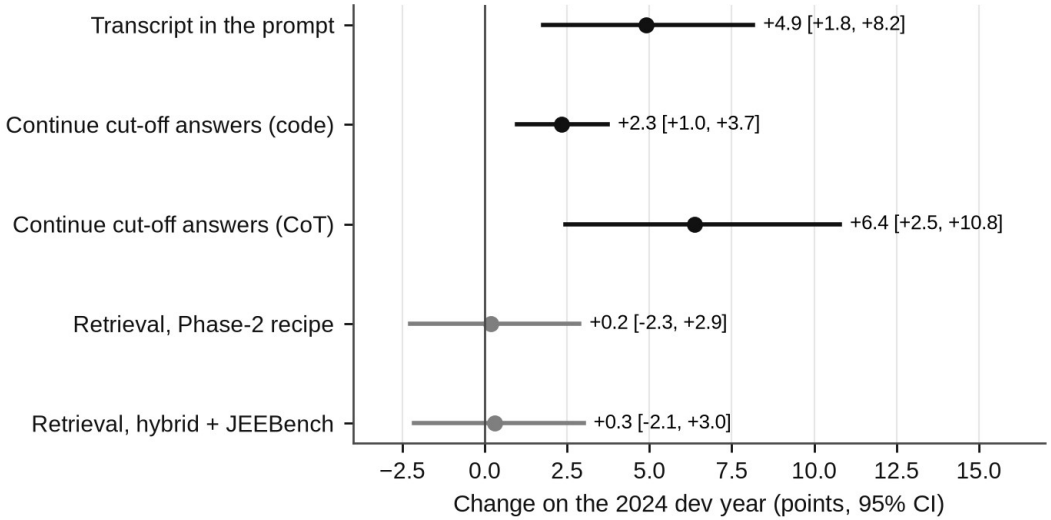


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HINDI · CROPPED BY US FROM THE OFFICIAL PAPER

- Held after Qwen3-VL was released, so the model cannot have seen it. The dataset had only the 102 English questions.
- We built the 102 Hindi twins: matched by paper, subject and number, answers from the English key, automatic checks on every pair, 14 crops checked by hand.

Every improvement was chosen on old exams only



2024 dev year, 204 questions. Rule fixed in advance: keep a change only if it gains and the lower end of its 95% interval is above zero (black = passed).

WHAT WENT INTO THE FINAL SYSTEM

Transcript

the model writes the question out in English first (Hindi gain +8.7)

Continuation

a cut-off answer continues once for 4,096 more tokens, in every row

Hybrid retrieval

tag + problem text, + JEEBench: coverage 8.4 → 11.7%

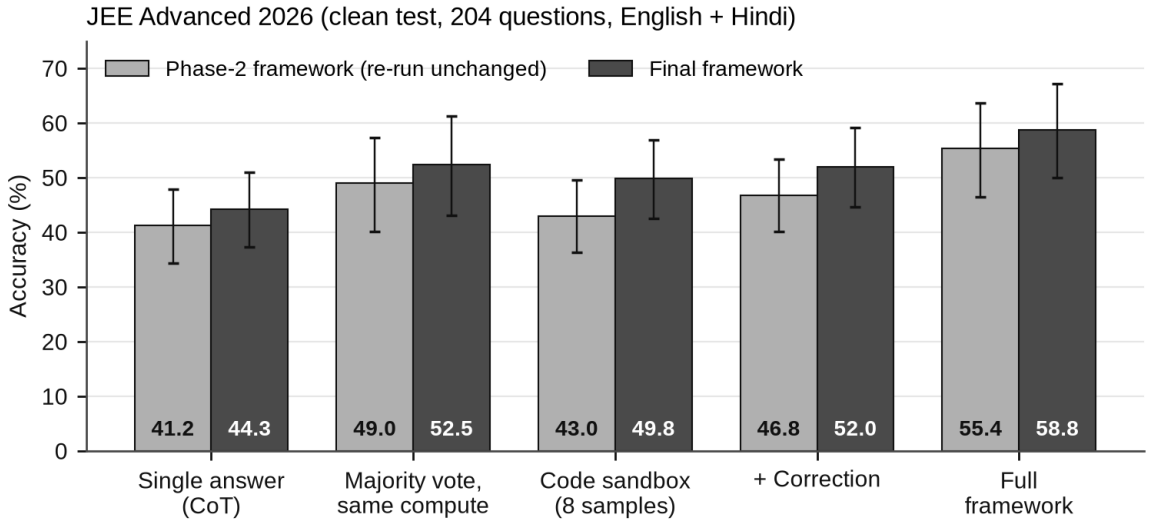
Ranking verifier

XGBoost ranker on pools with corrected answers: 62.0% vs. 53.7% for a vote

Correction gate

keep a fix only if the verifier score rises

58.8% on an exam the model has never seen



Bars with 95% intervals. Majority vote gets as many generated tokens as the full system (18 answers).

Our main claim holds on unseen data.

58.8%

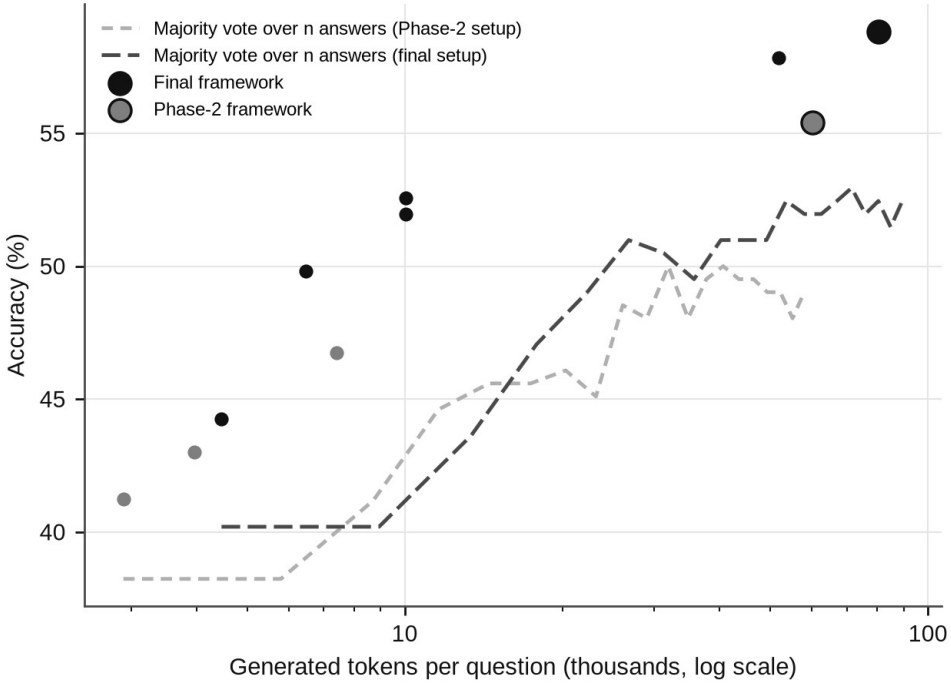
final system; one answer: 44.3%

+6.4

points over majority vote at equal compute
(+1.5 to +10.8, p = 0.011)

- The Phase-2 system, re-run unchanged, also beats its vote by +6.4 (p = 0.015).
- Stronger solver: +6.8 over the Phase-2 solver (p < 0.001).
- Final vs. Phase-2 system: +3.4, not significant on 204 questions.

The gain is not just more compute



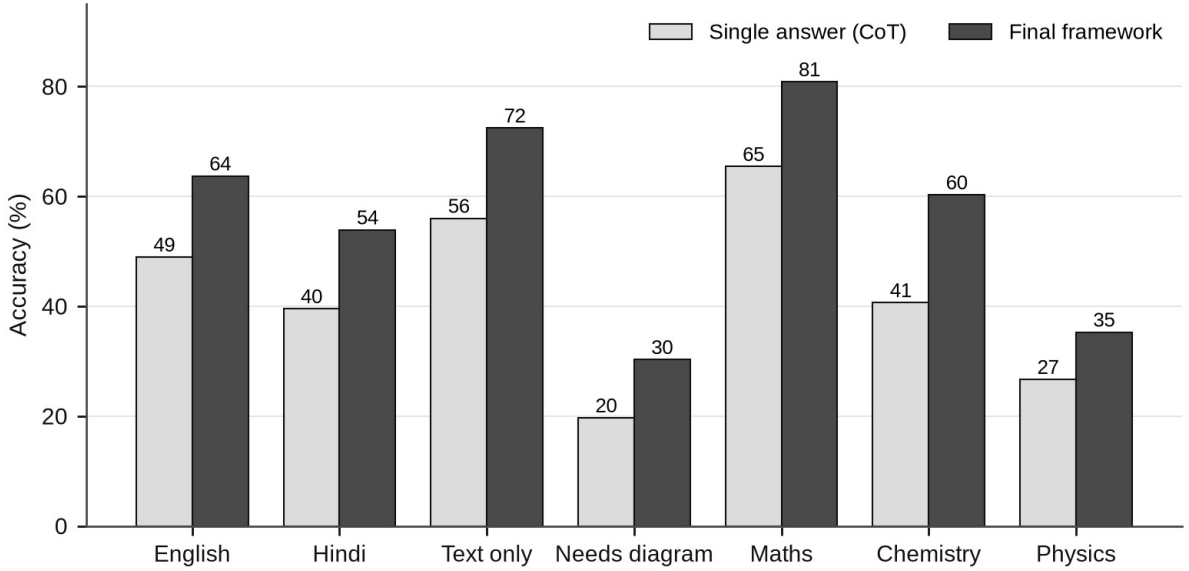
HOW TO READ IT

- Dashed lines: majority vote over n plain answers as n grows. This is what extra compute buys without our modules.
- Dots: our rows (small) and full systems (large). Every dot lies above the dashed line at its cost.

58.8%
final system,
80k tokens / question

52.5%
majority vote with
the same tokens

Gains everywhere, but diagrams stay hard



30% vs. 72%

diagram vs. text-only questions,
final system, 2026

- Physics, which has the most diagrams: 35%; Maths: 81%.
- Hindi 54% vs. English 64%; part of this may come from our own crops.
- Reasoning and code help; they do not fix reading the figure.

Correction still works on unseen data

Final system, 2026 (1,632 chains)	Value
Wrong answers flagged by the critic (round 1)	52.9%
Flagged errors fixed in round 1	13.2%
Correct answers wrongly flagged	19.9%
Accuracy per chain, before → after	49.8 → 52.0%

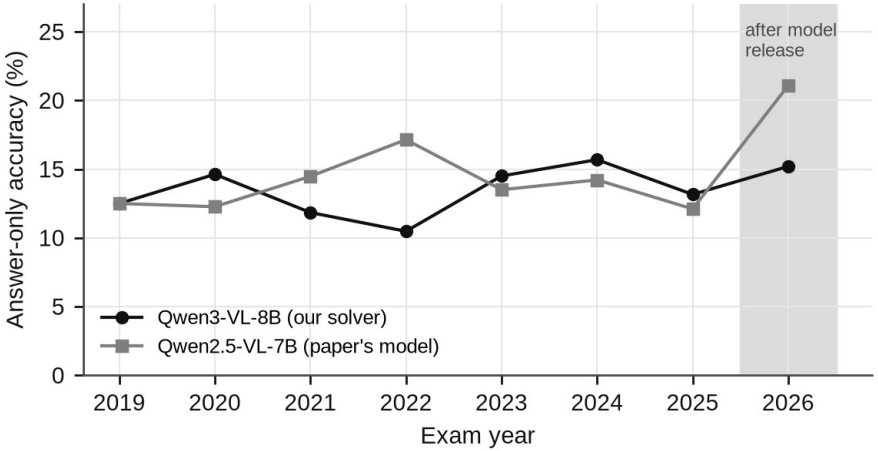
Base paper, single pass: 1.1–5.2% of detected errors fixed.

Correction gate, 2026	No gate	Gate
Right → wrong flips	**33**	**9**
Wrong → right flips	68	54
Accuracy per chain	52.0%	52.6%

The gate keeps a correction only if the verifier score rises.

> *Pointing at the first wrong step keeps working; the gate removes most harmful flips (33 → 9) but also blocks good fixes, so accuracy moves only +0.6.*

No memorisation, and the method needs a capable solver



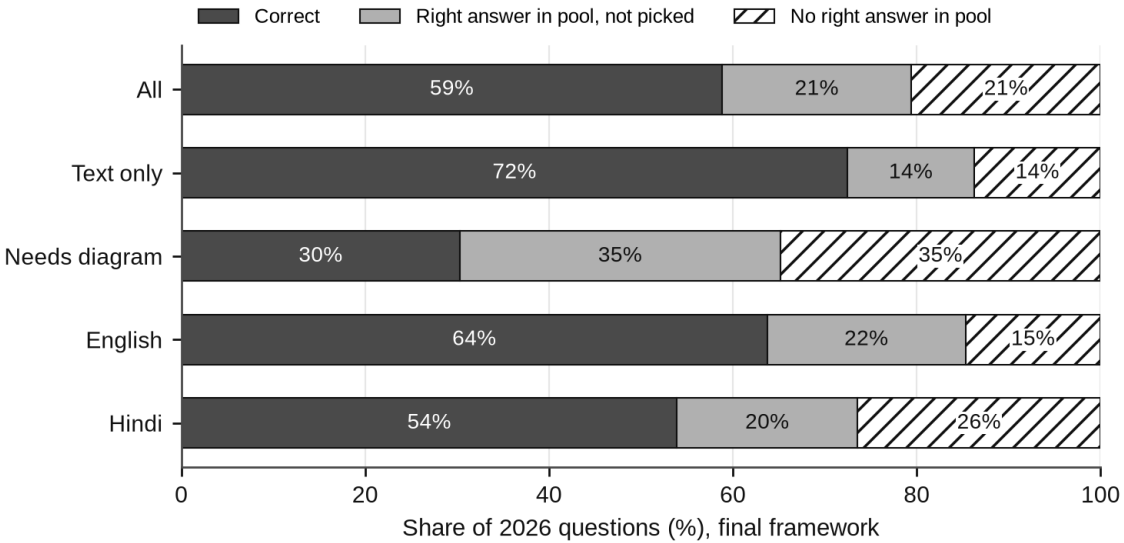
Answer-only probe (no reasoning). If old papers were memorised, old years would score higher than 2026. They do not.

WITH THE PAPER'S WEAKER MODEL (2025)

Qwen2.5-VL-7B	Accuracy
Single answer	14.0%
Majority vote, same compute	16.3%
Full framework	15.3%

- No gain: when only ~14% of answers are right, agreement, critique and verification carry little signal.
- 2025, used during design, shows bigger gains than 2026 (+13.7 vs. +6.4 over a vote), consistent with some optimism on a seen year.

Where the remaining errors are

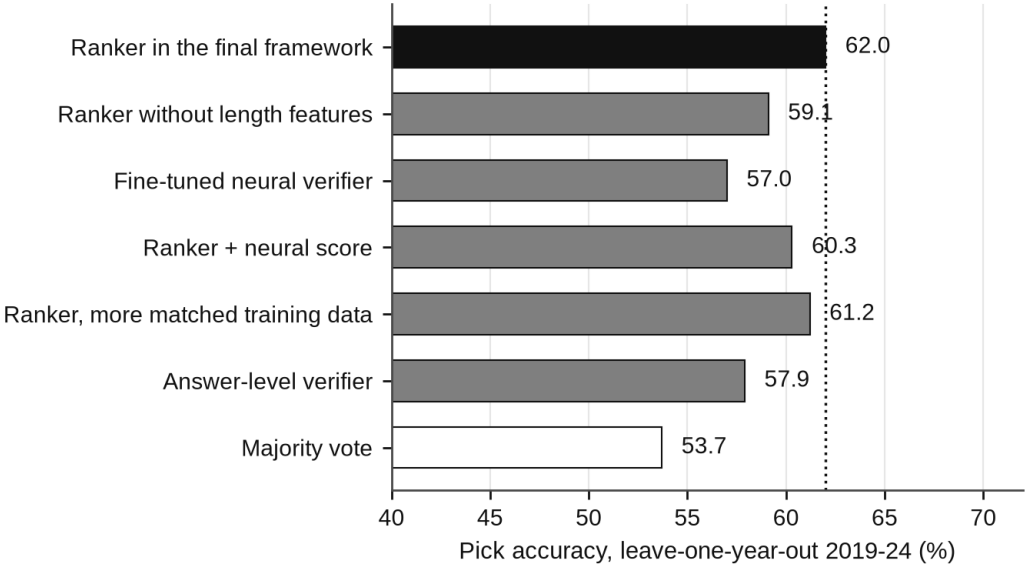


42
right answer in the pool, not picked

42
no right answer in the pool

- 120 of 204 questions solved.
- In the missed pools only 17% of answers are right: the right answer is a small minority.
- Diagram questions: 35% missed by the selector, 35% never solved.
- Longer answers are not the main cause: they are longer in both years, yet the verifier helps only on 2025.

Six fixes for selection and retrieval, none passed its rule



Pick accuracy, leave-one-year-out over 2019–2024 (no test data). Rule set in advance: beat the current ranker on every pool type. Ranker noise ≈ 1 point.

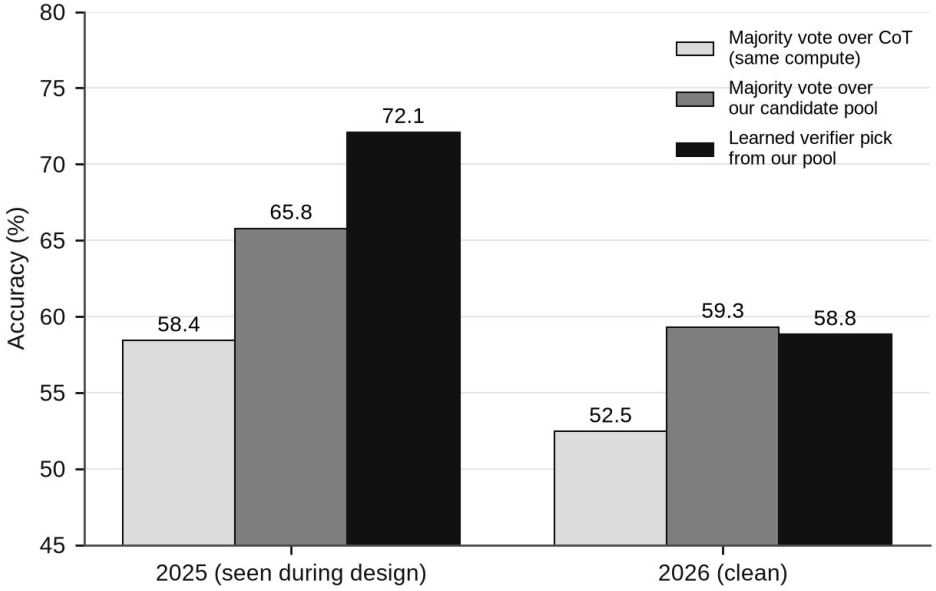
SELECTORS

- Fine-tuned neural verifier (ModernBERT): learns a signal (AUC 0.79), ranks worse.
- 3,184 new training answers matching the test pools (≈ 30 GPU-hours): 61.2%, within noise.

RETRIEVAL

- Method cards (method, plan, key equations, common mistake): +3.6 where retrieval fires (−0.8 to +8.5), so not adopted.
- 207 open-licensed OlympiadBench solutions: coverage 15.0 → 13.5%, 2026 accuracy unchanged.

The gain comes from the pool, not the picker



	2025	2026
Pool gain (vote on our pool − plain vote)	+7.4	+6.9
Selection gain (verifier pick − vote)	**+6.3**	**−0.5**
Verifier AUC, all answers	0.87	0.90
AUC inside mixed pools	0.80	0.75

The overall AUC rises on 2026 because it mostly separates easy questions from hard ones. Inside pools with both right and wrong answers, the only place picking matters, it falls.

> *A high AUC is not proof of good selection.*

Two last ideas, two clear answers

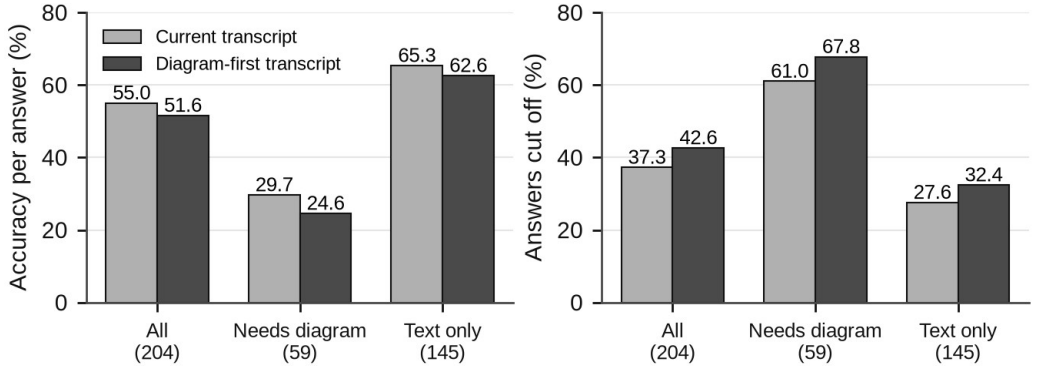
A · ANSWER-LEVEL VERIFIER

57.9%

vs. 62.0% for the current ranker
(worst year 50.0 vs. 52.5)

- One row per distinct answer, not per solution; pick the most likely answer.
- 5,335 answer rows from 15,280 solutions; model chosen by its worst year.
- Beats voting (53.7%) but loses the detail of single solutions.

B · DIAGRAM-FIRST TRANSCRIPT (2024 DEV)



- A second, itemised reading of the figure for half of each pool, same seeds.
- Majority vote −2.0 (−5.4 to +1.0); per answer −3.4, worst on diagram questions.
- Longer prompt, more cut-offs (37.3 → 42.6%): the solver misreads the figure whichever description it gets.

Everything we learned, in one table

Idea	Result	Why (from our analysis)
Code sandbox	numerical 40 → 50% (2025)	exact arithmetic instead of mental maths
Agentic correction	fixes 14–19% of detected errors	critic names one step; but flags 19–26% of right answers
Learned verifier	+11.8 on 2025; ±0 vs. vote on 2026	ranking inside mixed pools got worse on the new year
Retrieval	+0.2 to +2.1, not significant	fires for only 15–16% of questions
Transcript	+4.9 on dev (Hindi +8.7)	writing the question out helps reading, most of all in Hindi
Continue cut-off answers	+2.3 (code), +6.4 (plain) on dev	many answers were cut off at 4,096 tokens
Correction gate	bad flips 33 → 9, accuracy +0.6	blocks good fixes as well as bad ones
New selectors (4)	all below 62.0 (LOYO)	right answer is a small minority in missed pools
Method cards / open KB	+3.6 (not sure) / +0.0	too few questions where retrieval fires
Diagram-first reading	−2.0 on dev	longer prompt, more cut-offs; solver misreads figures
Weaker solver (7B)	no gain (15.3 vs. 16.3)	too few right answers for voting or checking

Running 8B models on one shared 12 GB GPU

Problem	What happened	What we did
Model did not fit	only 0.36 GB left for the attention cache	smaller images, 8-bit cache, batch limit: 1.9× faster
Runs killed silently	out-of-memory daemon killed a session (10.5 h lost)	detached supervisor, 3 retries per stage, quiet watcher
Three lab shutdowns	PC powered off; driver missing after reboot	driver installed, auto power-on; resumed with identical results
GPU crash at start	out of memory, then a CUDA error in a forked worker	embedding model on CPU; workers started with spawn
Second PC unusable	no GPU driver; campus proxy blocked downloads	packages copied over LAN; pre-compiled kernel installed
One prompt, whole batch lost	a 12,303-token prompt crashed hours of work	catch the error, retry request by request
Sandbox could eat RAM	a printing loop used 8 GB in a test	output written to size-capped files
No Hindi 2026 questions	the dataset had English only	cropped 102 Hindi questions from official papers

Optimisations that made it fit, and kept it honest

4-bit models

AWQ weights let two 8B models share one 12 GB GPU, one at a time.

Fast batched inference

vLLM batching, 8-bit attention cache, capped image size: 1.9× throughput.

Cache every call

each answer stored with a fixed seed: crashes and re-runs cost nothing; samples reused across experiments.

Cached embeddings

knowledge-base rebuild: 38 min → 1.8 min.

Matched seeds

with and without retrieval share seeds, so only questions where it fires can change.

Extend only what is cut off

instead of doubling every answer's budget (≈ 55 GPU-hours).

TESTING AND REPRODUCIBILITY

115	automated tests	11	bugs found in review before the first GPU run	3	freeze / pre-registration files, committed first	240+	hours of logged GPU runs

What this project achieved

+14.6

POINTS ON AN UNSEEN EXAM

44.3% → 58.8% on JEE Advanced 2026 with a frozen 8B model, no retraining.

+6.4

POINTS AT EQUAL COMPUTE

beats majority voting with the same tokens (p = 0.011), for both of our systems.

14–19%

OF DETECTED ERRORS FIXED

vs. 1.1–5.2% reported by the base paper for single-pass correction.

102

HINDI 2026 QUESTIONS BUILT

a clean bilingual test year that did not exist in the dataset.

8

IDEAS TESTED AND REPORTED

pre-registered negative results show what does not help, and why.

13.8%

VS. 12.4% REPRODUCED

the paper's own number, matched before we built on it; open repository with 115 tests.

Conclusion

On an exam the model cannot have seen, our framework lifts a frozen 8B model from 44.3% to 58.8% and beats majority voting at the same compute by 6.4 points.

The gain comes from a richer pool of candidate answers, and it is stable across years. Picking the best answer is the fragile part, and diagrams remain the main open problem.

LIMITATIONS

- one solver, one clean year of 204 questions (intervals ≈ ±8 points)
- Hindi 2026 crops are our own; results after the freeze are post-hoc

NEXT STEPS

- better perception for diagrams and Hindi, to put the right answer into more pools
- larger dev sets to settle method cards

> *Every claim was tested against a fair baseline, with rules fixed in advance, and every negative result is reported.*

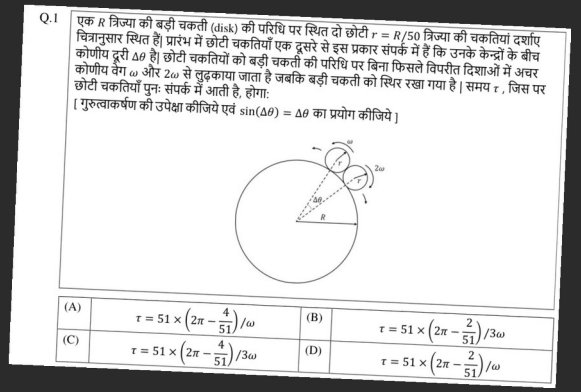
Thank you

Questions and discussion are welcome.

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Code, results and pre-registrations: github.com/Saurav1375/erdos-mmjee-reasoner



Q.1 एक R त्रिज्या की बड़ी चकती (disk) की परिधि पर स्थित दो छोटी $r = R/50$ त्रिज्या की चकतियां दर्शाए चित्रानुसार स्थित हैं| प्रारंभ में छोटी चकतियाँ एक दूसरे से इस प्रकार संपर्क में हैं कि उनके केन्द्रों के बीच कोणीय दूरी $\Delta\theta$ है| छोटी चकतियों को बड़ी चकती की परिधि पर बिना फिसले विपरीत दिशाओं में अचर कोणीय वेग ω और 2ω से लुढ़काया जाता है जबकि बड़ी चकती को स्थिर रखा गया है | समय τ , जिस पर छोटी चकतियाँ पुनः संपर्क में आती है, होगा:

[गुरुत्वाकर्षण की उपेक्षा कीजिये एवं $\sin(\Delta\theta) = \Delta\theta$ का प्रयोग कीजिये]

(A)	$\tau = 51 \times \left(2\pi - \frac{4}{51}\right)/\omega$	(B)	$\tau = 51 \times \left(2\pi - \frac{2}{51}\right)/3\omega$
(C)	$\tau = 51 \times \left(2\pi - \frac{4}{51}\right)/3\omega$	(D)	$\tau = 51 \times \left(2\pi - \frac{2}{51}\right)/\omega$

The Hindi twin of the title slide's question, cropped by us from the official 2026 paper.